# Curriculum Vitae

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## Academic appointments

from 08/2026 **Postdoctoral researcher**, *ETH Zürich*, Mentor: Prof. Dr Mikaela Iacobelli

04/2023–06/2026 **Postdoctoral researcher**, *University of Münster*, Mentor: Prof. Dr Christian Seis

04/2019–03/2023 **Research associate**, *Ulm University*

## Education

04/2019–06/2023 **PhD in Mathematics**, *Ulm University*
Thesis: Analytic aspects of kinetic partial differential equations
Advisor: Prof. Dr Rico Zacher
Referees: Prof. Dr Cyril Imbert, Prof. Dr Hongjie Dong
Grade: summa cum laude; recipient of the Ulmer Universitätsgesellschaft's PhD thesis award

03/2018–03/2019 **MSc in Mathematics**, *Ulm University*
Thesis: Kolmogorov equations – Well-posedness, regularity, asymptotics and Harnack inequalities
Grade: 1.0 with distinction; received the award for the best graduating student

10/2014–02/2018 **BSc in Mathematics**, *Ulm University*
Thesis: Long-time behaviour of Markov chains by discrete functional inequalities and entropic Ricci curvature
Grade: 1.1 with distinction; received the award for the best graduating student

04/2013–09/2014 **Junior studies in mathematics**, *FernUniversität in Hagen and PFH Göttingen*

06/2014 **Abitur**, *Joachim-Hahn-Gymnasium Blaubeuren*

## Awards and Scholarships

06/2026 DFG Walter Benjamin Fellowship for a research project at the University of Cambridge, hosted by Prof. Clément Mouhot (approximately 100,000 EUR, declined)

02/2025 PhD thesis award "Promotionspreis der Ulmer Universitätsgesellschaft" (1500 EUR)

06/2022 Travel grant from the ProTrainU Mobility-Programme, Ulm University (1000 EUR)

04/2022 Master's graduation award "Absolventenpreis M.Sc. Mathematik Uni Ulm 2020" for the best graduating student (500 EUR)

11/2019 Graduate scholarship granted by the State of Baden-Wuerttemberg, Germany (52,200 EUR)

05/2018 Bachelor's graduation award "Absolventenpreis B.Sc. Mathematik Uni Ulm 2018" for the best graduating student (500 EUR)

06/2014 Ferry Porsche award and graduation award of the Deutsche Physikalische Gesellschaft

## Research

### Preprints

[1] **An Aleksandrov estimate for kinetic equations with rough coefficients** with Amélie Loher, Connor Mooney, Clément Mouhot, Luis Silvestre, and Yuzhe Zhu. arXiv:2610.13921, 2026.

[2] **Existence and upper bounds of fundamental solutions to kinetic equations with rough non-local diffusion** with Pascal Auscher. arXiv:2610.06214, 2026.

[3] **On Leonardo da Vinci's paradox** with Björn Gebhard and Christian Seis. arXiv:2609.26283, 2026.

[4] **A Liouville theorem for the two-dimensional stationary hypodissipative Navier–Stokes system** with Nicola De Nitti and Jiaqi Yang. arXiv:2608.16437, 2026.

[5] **Lions' maximal regularity problem for divergence-form differential operators: failure at the $\frac{1}{2}$-Hölder endpoint**. arXiv:2608.11194, 2026.

[6] **A fast dynamo with zero topological entropy**. arXiv:2608.02586, 2026.

[7] **Sharp kinetic trace theory** with Lisa Valentini. arXiv:2607.24708, 2026.

[8] **On the kinetic $p$-Laplace equation with nonlocal diffusion**. arXiv:2605.21080, 2026.

[9] **Nonlinear kinetic diffusion equations with $p$-growth** with Helge Dietert and Rico Zacher. arXiv:2605.18521, 2026.

[10] **Transfer of regularity by kinetic mollification along critical trajectories**. arXiv:2605.13582, 2026.

[11] **Spherical rigidity for an exterior overdetermined problem with Neumann data prescribed by mean curvature**. arXiv:2604.07002, 2026.

[12] **Kinetic Sobolev Spaces** with Pascal Auscher. arXiv:2603.17491, 2026.

[13] **Nash's $G$ bound for the Kolmogorov equation** with Helge Dietert. arXiv:2510.21621, 2025.

[14] **Critical trajectories in kinetic geometry** with Helge Dietert, Clément Mouhot, and Rico Zacher. arXiv:2508.14868, 2025.

[15] **Weak solutions to Kolmogorov–Fokker–Planck equations: regularity, existence and uniqueness** with Pascal Auscher and Cyril Imbert. arXiv:2403.17464, 2024.

### Publications

[16] **Global rigidity of two-dimensional bubbles**. *Arch. Ration. Mech. Anal.*, 250:Article 104, 2026. [DOI].

[17] **Steady bubbles and drops in inviscid fluids** with David Meyer and Christian Seis. *Calc. Var.*, 64(299), 2025. [DOI].

[18] **A trajectorial interpretation of Moser's proof of the Harnack inequality** with Rico Zacher. *Ann. Sc. Norm. Super. Pisa Cl. Sci. (5)*, 26(2):605–617, 2025. [DOI].

[19] **Fundamental solutions to Kolmogorov–Fokker–Planck equations with rough coefficients: existence, uniqueness, upper estimates** with Pascal Auscher and Cyril Imbert. *SIAM J. Math. Anal.*, 57(2):2114–2137, 2025. [DOI].

[20] **On a kinetic Poincaré inequality and beyond** with Rico Zacher. *J. Funct. Anal.*, 289(1):Paper No. 110899, 18, 2025. [DOI].

[21] **Analytic aspects of kinetic partial differential equations**. *OPARU — Open Access Repositorium der Universität Ulm*, 2023. [DOI].

[22] **Kinetic maximal $\mathrm{L}^p$-regularity with temporal weights and application to quasilinear kinetic diffusion equations** with Rico Zacher. *J. Differential Equations*, 307:29–82, 2022. [DOI].

[23] **Kinetic maximal $\mathrm{L}^p_\mu(\mathrm{L}^p)$-regularity for the fractional Kolmogorov equation with variable density**. *Nonlinear Anal.*, 214:Paper No. 112517, 21, 2022. [DOI].

[24] **Kinetic maximal $\mathrm{L}^2$-regularity for the (fractional) Kolmogorov equation** with Rico Zacher. *J. Evol. Equ.*, 21(3):3585–3612, 2021. [DOI].

## Employment History

10/2022–03/2023 **Predoctoral researcher**, *Ulm University*

03/2020–10/2022 **Teaching assistant**, *University of Applied Sciences Neu-Ulm*

10/2015–10/2019 **Teaching assistant**, *Ulm University*

03/2015–03/2019 **Research assistant (working student)**, *Daimler AG Research & Development Ulm*

## Teaching Experience

### ETH Zürich

Autumn 2026 Instructor for the student seminar *Collisional Kinetic Theory: Entropy and Fisher Information*

Supervision of one semester project

### University of Münster

2023–2025 Teaching assistant for *Partial Differential Equations* (1 sem.) and *Analysis and Numerics of Differential Equations* (1 sem.) (exercise-sheet design; large-class solution sessions; tutor and marking coordination; exam support)

Winter 2024/25 Supervised a student seminar on *De Giorgi–Nash–Moser Theory*

2024–2025 Examined two BSc theses; co-supervised and examined the MSc thesis of Yuanjiang Han (now a PhD student with Christian Seis)

### University of Applied Sciences Neu-Ulm

2020–2022 Teaching assistant for *Business Mathematics and Statistics I* (3 sem.) and *Business Mathematics and Statistics II – Regression Models / Statistical Testing* (2 sem.) (authored exercise sheets; ran the programming lab; marked submissions; created teaching videos)

### Ulm University

2017–2022 Teaching assistant for *Analysis I* (2 sem.), *Analysis II* (1 sem.), *Dynamical Systems* (2 sem.), and *Elements of Functional Analysis* (1 sem.) (exercise-sheet design; large-class solution sessions; tutor and marking coordination; exam-setting support)

Winter 2022/23 Local coordinator of the 26th Internetseminar

Winter 2019/20 Organised the "Trainingscamp", an annual four-week maths skills refresher for first-year students

2015–2017 MathLab tutor for *Analysis I* and *Analysis II*

## Talks and Posters

08/2026 Talk – Nonlinear PDEs, Harmonic Analysis and Free Boundary Problems, KTH Stockholm

05/2026 Talk – Calderón–Zygmund Analysis Seminar, University of Chicago

05/2026 Talk – Applied and Computational Mathematics Seminar, Tulane University

04/2026 Talk – ISU Analysis and Probability Seminar, Iowa State University

03/2026 Talk – Horizons in nonlinear PDEs, Ulm University

10/2025 Talk – Oberseminar Angewandte Analysis, Dortmund University

10/2025 Talk – Noko25 43rd Northern German Colloquium, Humboldt-Universität zu Berlin

09/2025 Poster – Hyperbolic Problems in Nürnberg 2025, FAU Erlangen

05/2025 Talk – Seminar Nichtlineare Partielle Differentialgleichungen, KIT Karlsruhe

03/2025 Poster – Kinetic equations and turbulence, CentraleSupélec and Collège de France

10/2024 Talk – Münster–Imperial Day in PDE, University of Münster

09/2024 Talk – Oberseminar Analysis, University of Bonn

03/2024 Poster – Mathematics Münster Mid-term Conference, University of Münster

11/2023 Talk – Séminaire Analyse Numérique et EDP, Université Paris-Saclay

09/2023 Talk – Seminar of the AG Analysis-Probability, MPI Leipzig

06/2023 Talk – Real Analysis and Geometry, CIRM Marseille

04/2023 Talk – Wilhelm Killing Colloquium, University of Münster

03/2023 Talk – Chair Dynamics Control and Numerics, FAU Erlangen

02/2023 Talk – TULKKA, KIT Karlsruhe

11/2022 Poster – Kinetic Theory, CIRM Marseille

09/2022 Talk & Poster – Horizons in non-linear PDEs, Ulm University

07/2022 Poster – When Kinetic Theory meets Fluid Mechanics, ETH Zürich

06/2022 Poster – 11th Summer school on "Methods and Models of Kinetic Theory", Pesaro

01/2022 Poster – Frontiers in analysis of kinetic equations, INI Cambridge

07/2021 Talk – Nonlinear Evolution Equations and Approximations, University of Duisburg-Essen

03/2021 Talk – Oberseminar Analysis, Ulm University

02/2021 Poster – Winterschool on Analysis and Applied Mathematics, University of Münster

## Research Stays

05/2026 Visit to Luis Silvestre (University of Chicago), one week

04/2026 Visit to Sam Punshon-Smith (Tulane University), one week

04/2026 Visit to David Herzog (Iowa State University), two weeks

2023–2025 Multiple visits to Pascal Auscher (Université Paris-Saclay), totalling 7 weeks

2023–2025 Multiple visits to Rico Zacher (Ulm University), totalling 5 weeks

2021–2025 Three visits to Cyril Imbert (École normale supérieure, Paris), totalling two weeks

05/2025 Visit to Björn de Rijk (KIT Karlsruhe), one week

09/2023 Visit to Tobias Ried (MPI Leipzig), one week

03/2023 Visit to Nicola De Nitti (FAU Erlangen), one week

02/2023 Visit to Christian Seis (University of Münster), half a week

## Other Scientific Activities

2026 *We need ABP*, an open mathematical collaboration on Aleksandrov–Bakelman–Pucci estimates, combining AI exploration with mathematical analysis and exposition of central ideas. `https://weneedabp.github.io/`

2024 & 2026 Supporting the organisation of the Münster–Imperial Day in PDE, University of Münster

2022 Supporting the organisation of the summer school "Horizons in non-linear PDEs" at Ulm University

2020 Co-organisation of the seminar "Mathematisches Kolloquium" at Ulm University for six semesters

2022 Organisation of a weekend workshop for mathematics PhD students at Ulm University: 15 participants, secured 2500 EUR in funding from StuVe Ulm

2022 Supervision of a three-month project for the workshop of the 25th Internetseminar. Title: Kolmogorov equation in $L^p(\mathbb{R}^{2n})$ – Semigroup, spectrum, maximal regularity?

2021 Facilitator at the autumn school "COLLAB – Collaboratory for Global Sustainability 2021", Toronto/Ulm

2019 Supporting the organisation of the winter school "Grad. Flows and Var. Methods in PDEs" at Ulm University

Active contribution to the PDE seminars at the University of Münster and Ulm University by inviting several external speakers

Attended over 30 international workshops and conferences, including events at CIRM Marseille, INI Cambridge, ETH Zürich, and the University of Oxford

Peer reviewer for *Annales scientifiques de l'École normale supérieure*, *Communications in Partial Differential Equations*, *Discrete and Continuous Dynamical Systems*, *European Journal of Mathematics*, *Journal of Evolution Equations*, *Kinetic and Related Models*, *Mathematika*, *Nonlinearity* and *Potential Analysis*

## Academic Volunteer Work

02/2021–09/2022 **Elected member of the council of PhD students**, *Ulm University*
Representing the interests of doctoral students in university committees, e.g. during revisions to the PhD regulations; chair of the council of PhD students (12/2021–09/2022)

02/2021–09/2022 **Elected member of the faculty council of the Faculty of Mathematics and Economics**, *Ulm University*

07/2017–03/2019 **Member of the Mathematics Student Council**, *Ulm University*
Member of the appointment committee for a professorship in "Stochastics"